Missing Numbers

AlgoExpert · Medium · Arrays

The problem

- Array holds 1..n+2 with two numbers missing; return them ascending.
- Split by the average of the two missing values and sum each half.

Intuition

- The numbers one to n plus two sum to a known total; the missing pair splits around their average.

Approach

- 1. Compute the missing sum
- 2. Split at the average
- 3. Sum the expected and actual values below it
- 4. The difference gives one number, the rest gives the other

Java solution

```java
public class MissingNumbers {
    public static int[] missingNumbers(int[] nums) {
        int n = nums.length + 2;
        long total = (long) n * (n + 1) / 2;
        for (int x : nums) total -= x;
        long avg = total / 2, lowExpected = avg * (avg + 1) / 2, lowActual = 0;
        for (int x : nums) if (x <= avg) lowActual += x;
        int a = (int) (lowExpected - lowActual);
        return new int[]{a, (int) (total - a)};
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- The numbers one to n plus two sum to a known total; the missing pair splits around their average.
- Complexity: O(n)